Pooja Kiran

Security Engineer | AI & Agent Security | Application Security | Cloud IAM Security | Model Supply-Chain Security

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Experience

Independent AI Security Researcher & Engineer Aug. 2024 - Present

Self-Directed Research *Tempe, AZ, USA*

- Mapped agent-to-tool, cloud IAM, and model supply-chain threats to enforceable controls, static checks, and repeatable regression tests.
- Automated Python regression and GitHub Actions validation across 3 self-directed security projects; dated CI evidence includes **782 gateway tests**, **25 IAM rule IDs**, and **260 model-scanner tests**.

Business & Compliance Lead, AEROSEC (academic externship) Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University *Tempe, AZ, USA*

- Developed compliance and commercialization analysis for a proposed airline PSS security layer, including a **$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University *Mesa, AZ*

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

Projects

Project periods are candidate-reported; metrics cite dated 2026 CI snapshots.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security* Oct. 2025 - Sep. 2026

- Built a default-deny MCP/JSON-RPC 2.0 gateway with 5 pre-execution controls: capability authorization, prompt-injection inspection, PII/exfiltration checks, anti-SSRF egress policy, and rate limiting.
- Expanded the prompt-injection collection from **55 earlier patterns to 69 current patterns**, with tamper-evident policy-decision audit logging.
- Cited Python 3.12 CI: **782 passing tests** at **81.93% statement coverage** in the same run. Includes **9 Elastic rules** and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0* Apr. 2025 - Sep. 2025

- Built an IAM static analyzer with **25 deterministic rules** covering wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries.
- Produced 3 finding formats (text, JSON, SARIF 2.1.0), integrating SARIF with GitHub Code Scanning for reviewable IAM policy risks.
- Verified 2026 CI snapshot: **247 collected, 244 passed, 3 skipped**; the skipped checks require live AWS credentials.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras* Nov. 2024 - Mar. 2025

- Engineered non-executing inspection across 5 model artifact families: SafeTensors, GGUF, ONNX, Keras, and pickle-derived files; avoided running untrusted model code by default.
- Combined pickle-opcode, AST/taint, provenance/dependency, and obfuscation checks; detected 33/33 committed adversarial fixtures (12 incident recreations, 18 variants, 3 large-scale cases).
- Verified 2026 CI snapshot: **260 passed, 1 skipped** with **75.67% statement coverage**; fixture detection separately evaluated.

Education

Arizona State University Tempe, AZ, USA

*M.S., Information Technology (Security), GPA: **3.87/4.00*** *Aug. 2024 - May 2026*

- **Academic Participation:** Student participant, ASU, 2025; Generative AI learning assistant for personalized education.

M. S. Ramaiah University of Applied Sciences Bengaluru, KA, India

*B.Tech., Computer Science & Engineering, CGPA: **8.44/10*** *Aug. 2019 - Aug. 2023*

Technical Skills

AI & Agent Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security

Identity & Application Security: Cloud IAM Security, AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Capability Authorization, Threat Modeling, API Security

Security Engineering & DevSecOps: Python, FastAPI, pytest, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, Docker

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Tamper-Evident Audit Logging, MITRE ATLAS

Certifications

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)